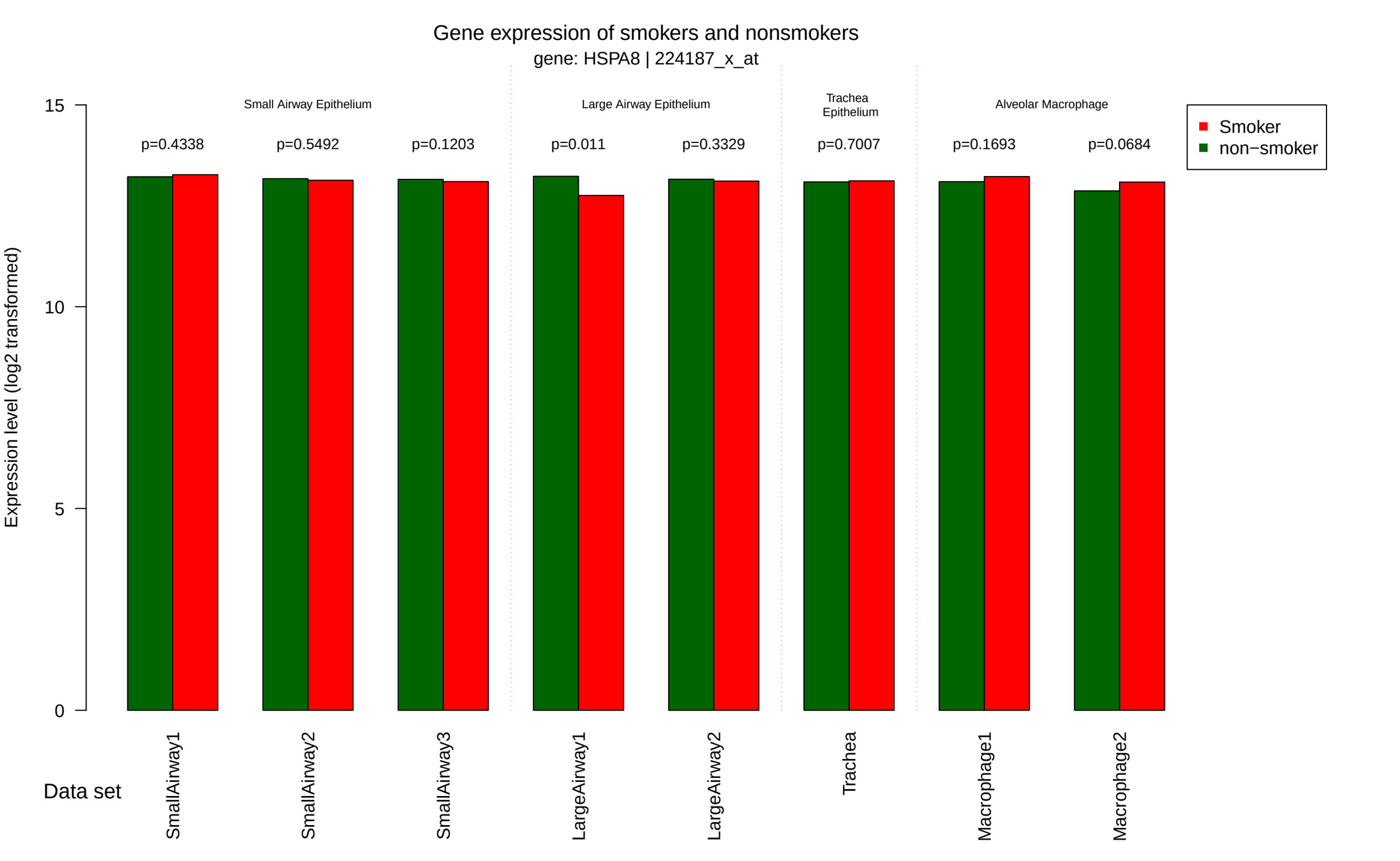
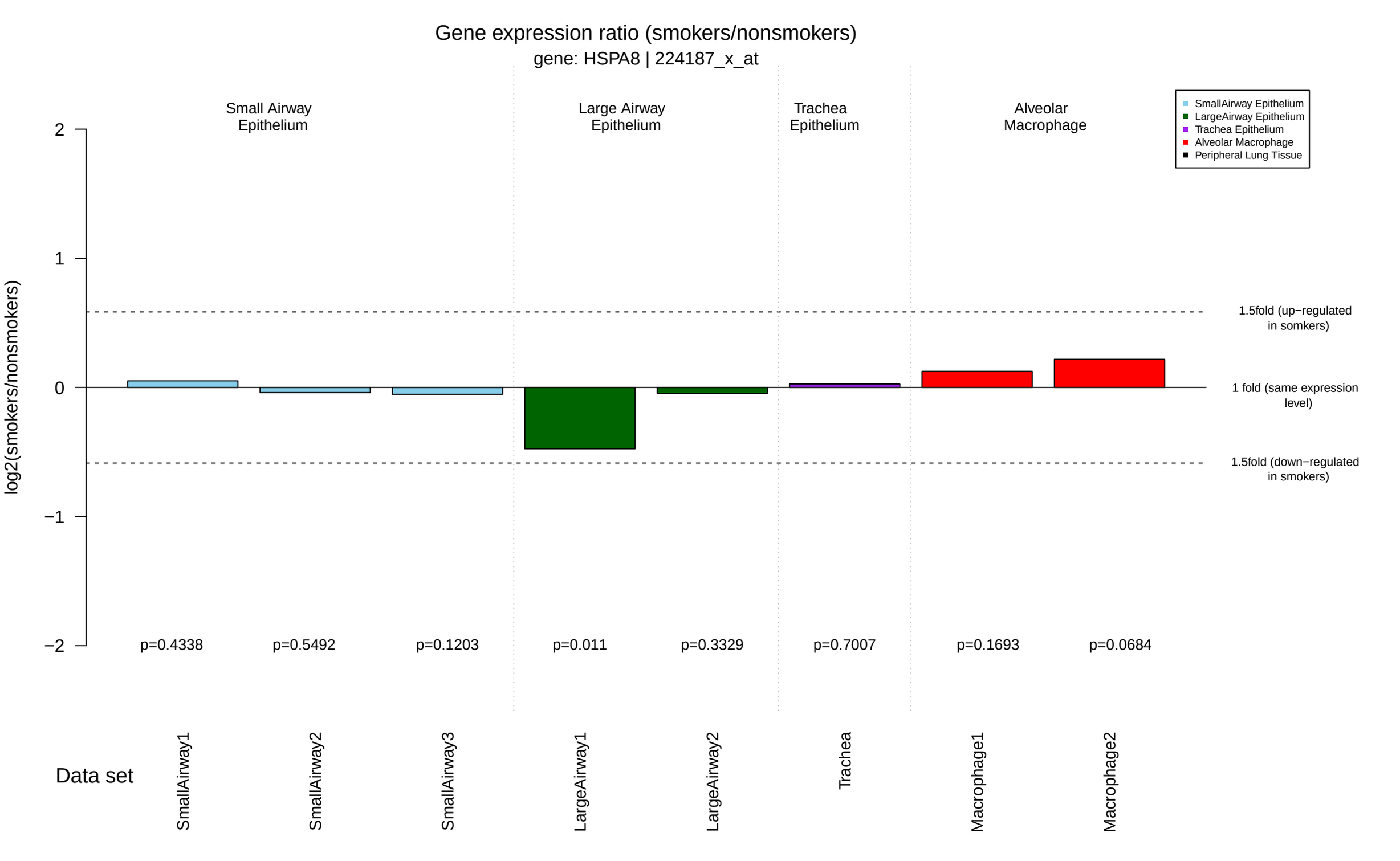
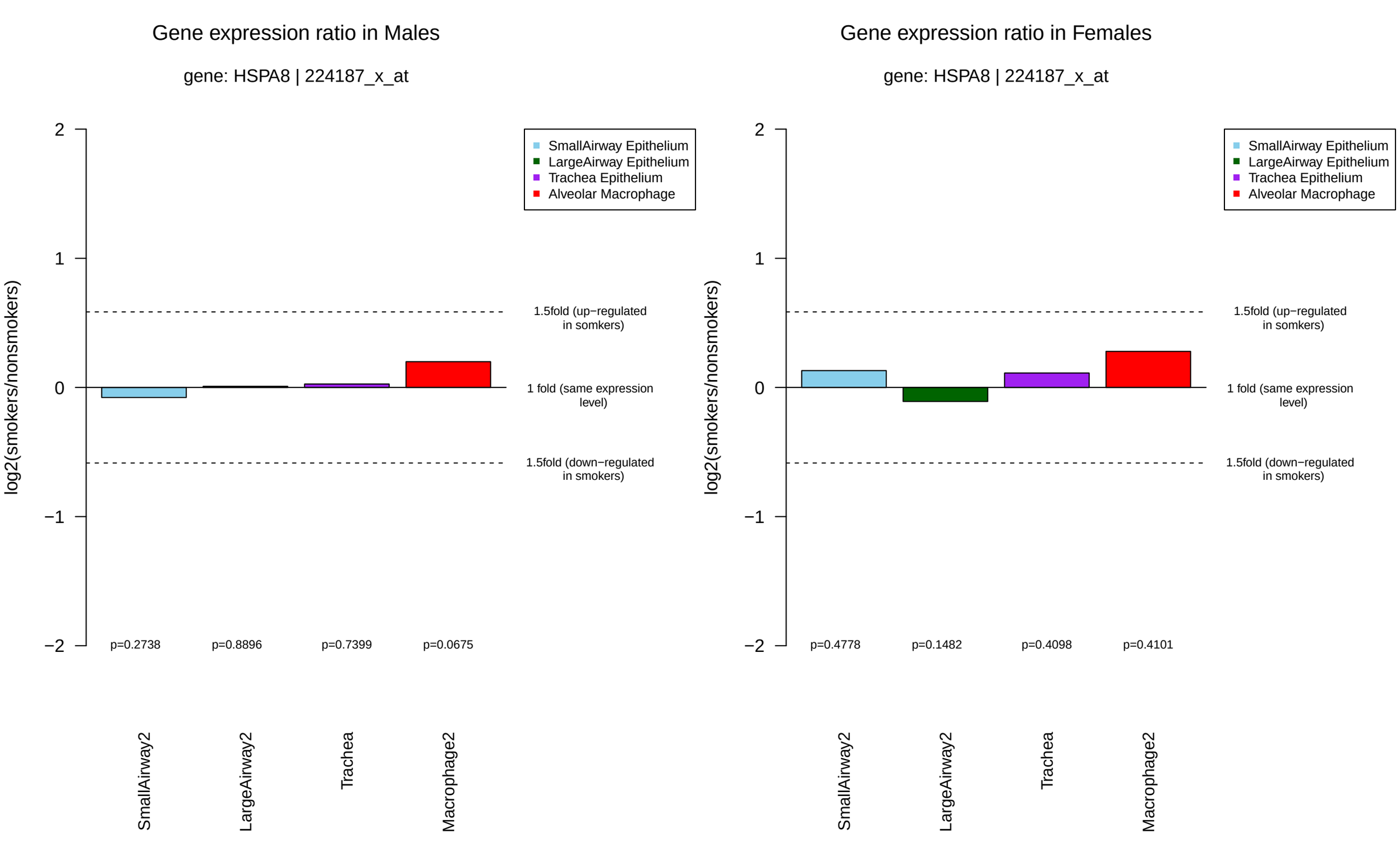
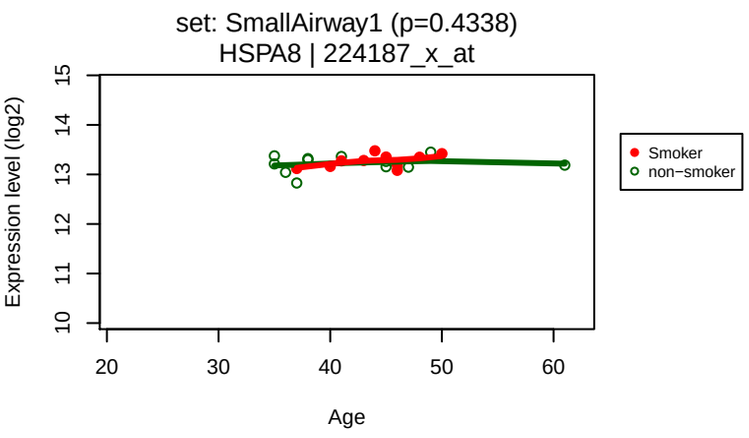
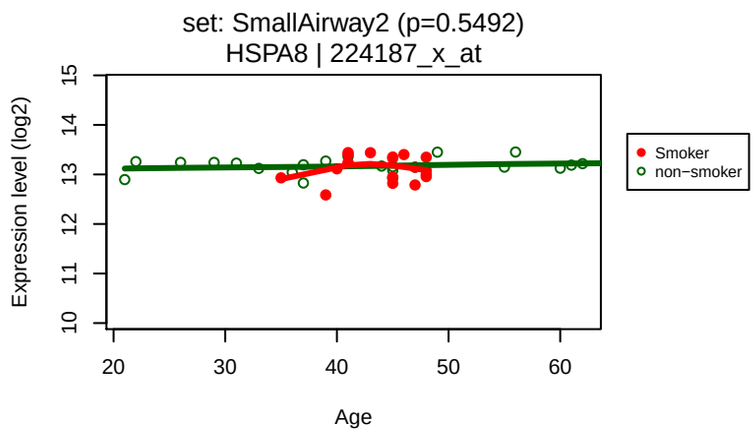
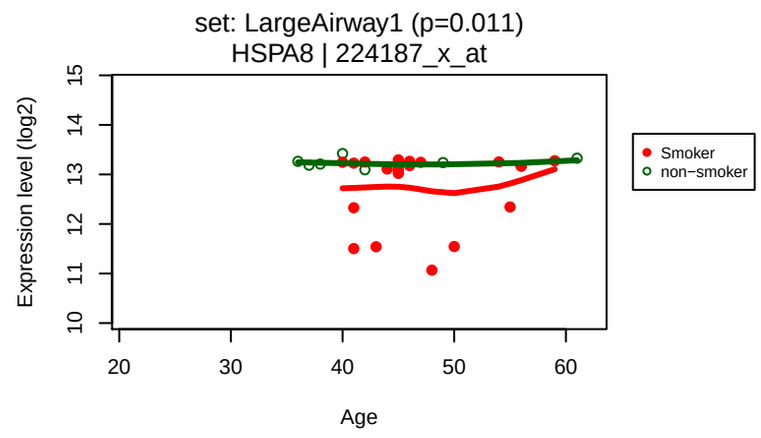

Gene expression of smokers and nonsmokers
gene: HSPA8 | 224187_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4338
p=0.5492
p=0.1203
p=0.011
p=0.3329
p=0.7007
p=0.1693
p=0.0684
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

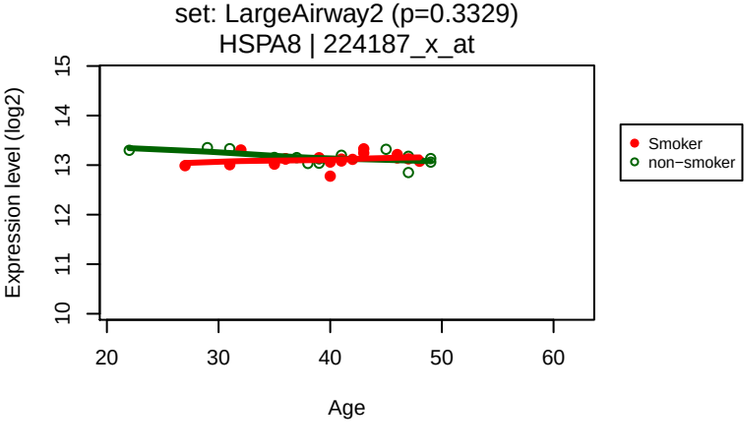

Gene expression ratio (smokers/nonsmokers)
gene: HSPA8 | 224187_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4338
p=0.5492
p=0.1203
p=0.011
p=0.3329
p=0.7007
p=0.1693
p=0.0684
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

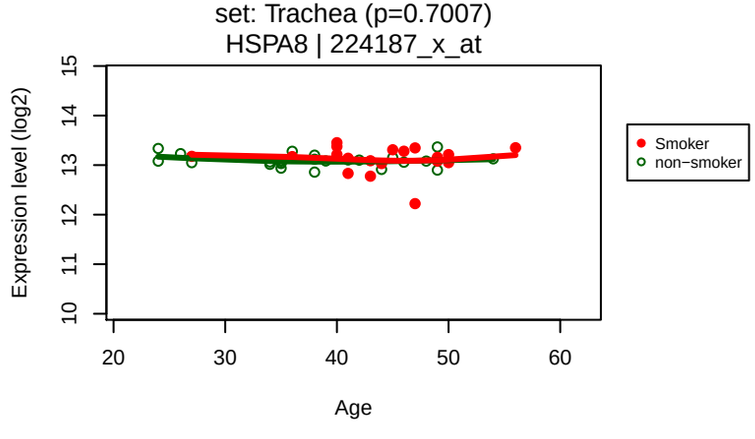

Gene expression ratio in Males
gene: HSPA8 | 224187_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2738
p=0.8896
p=0.7399
p=0.0675
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPA8 | 224187_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4778
p=0.1482
p=0.4098
p=0.4101
SmallAirway2
LargeAirway2
Trachea
Macrophage2

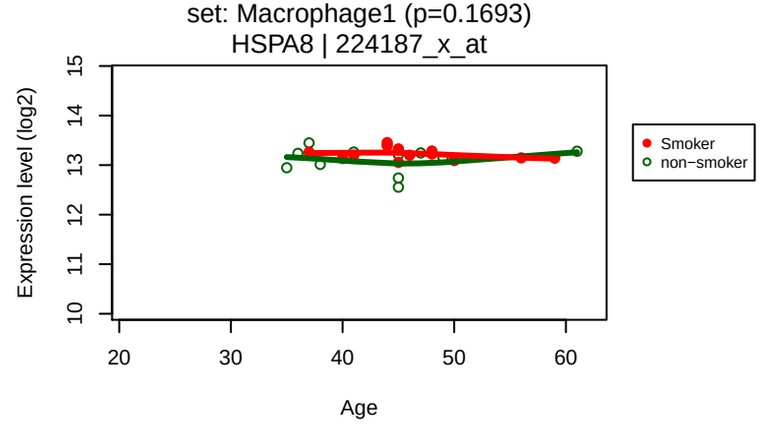

set: SmallAirway1 (p=0.4338)
HSPA8 | 224187_x_at
Expression level (log2)
Age
Smoker
non−smoker

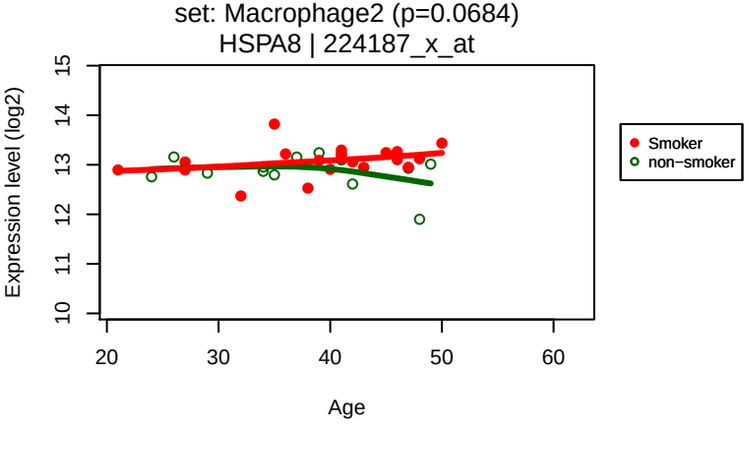

set: SmallAirway2 (p=0.5492)
HSPA8 | 224187_x_at
Expression level (log2)
Age
Smoker
non−smoker

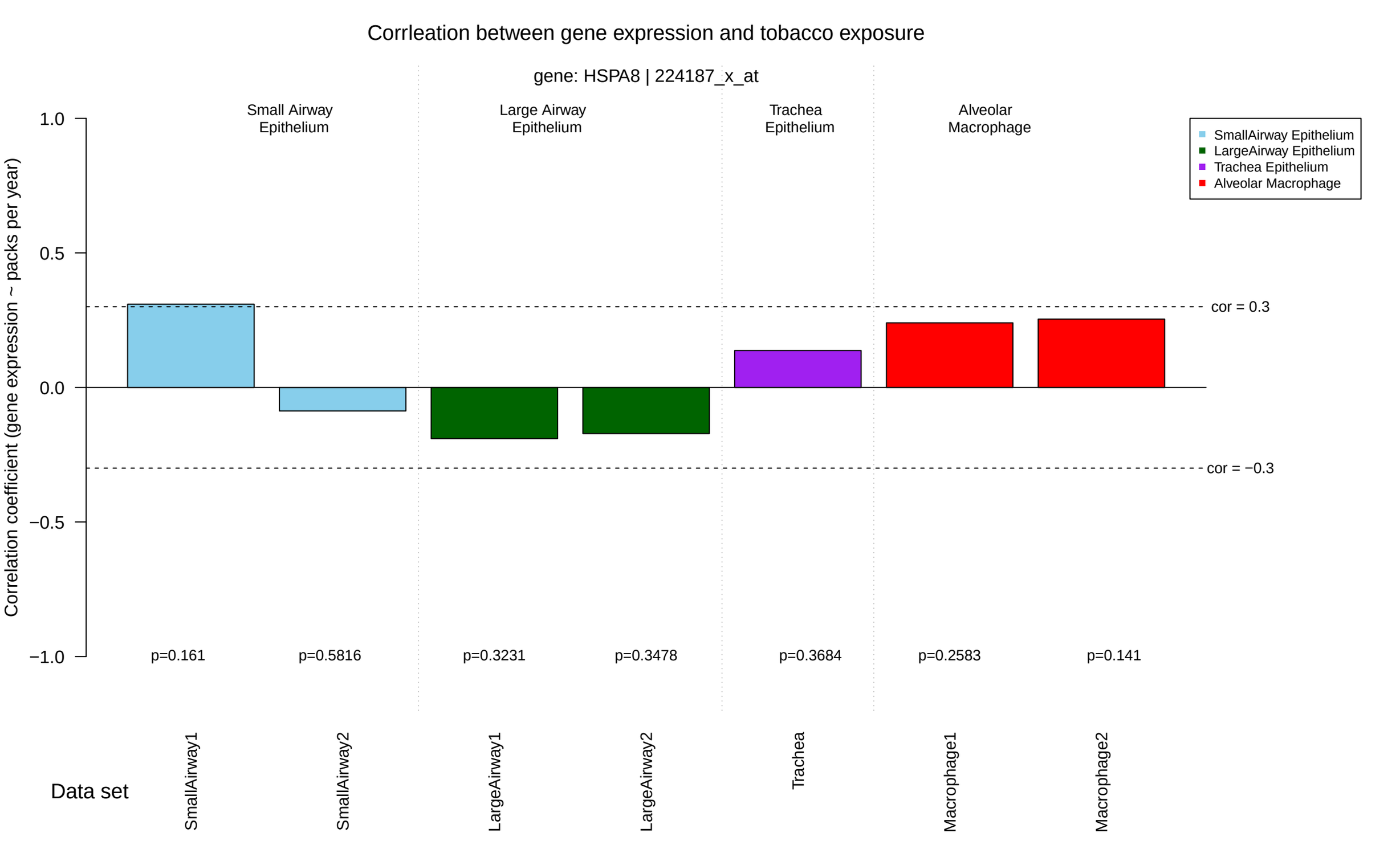

set: LargeAirway1 (p=0.011)
HSPA8 | 224187_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3329)
HSPA8 | 224187_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7007)
HSPA8 | 224187_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1693)
HSPA8 | 224187_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0684)
HSPA8 | 224187_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPA8 | 224187_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.161
p=0.5816
p=0.3231
p=0.3478
p=0.3684
p=0.2583
p=0.141
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2